
Contents

<i>Prologue</i>	ix
<i>Acknowledgments</i>	xii
1 Empirical Bayes and the James–Stein Estimator	1
1.1 Bayes Rule and Multivariate Normal Estimation	2
1.2 Empirical Bayes Estimation	4
1.3 Estimating the Individual Components	7
1.4 Learning from the Experience of Others	10
1.5 Empirical Bayes Confidence Intervals	12
Notes	14
2 Large-Scale Hypothesis Testing	15
2.1 A Microarray Example	15
2.2 Bayesian Approach	17
2.3 Empirical Bayes Estimates	20
2.4 $\text{Fdr}(\mathcal{Z})$ as a Point Estimate	22
2.5 Independence versus Correlation	26
2.6 Learning from the Experience of Others II	27
Notes	28
3 Significance Testing Algorithms	30
3.1 p -Values and z -Values	31
3.2 Adjusted p -Values and the FWER	34
3.3 Stepwise Algorithms	37
3.4 Permutation Algorithms	39
3.5 Other Control Criteria	43
Notes	45
4 False Discovery Rate Control	46
4.1 True and False Discoveries	46
4.2 Benjamini and Hochberg’s FDR Control Algorithm	48
4.3 Empirical Bayes Interpretation	52

4.4	Is FDR Control “Hypothesis Testing”?	58
4.5	Variations on the Benjamini–Hochberg Algorithm	59
4.6	Fdr and Simultaneous Tests of Correlation	64
	Notes	69
5	Local False Discovery Rates	70
5.1	Estimating the Local False Discovery Rate	70
5.2	Poisson Regression Estimates for $f(z)$	74
5.3	Inference and Local False Discovery Rates	77
5.4	Power Diagnostics	83
	Notes	88
6	Theoretical, Permutation, and Empirical Null Distributions	89
6.1	Four Examples	90
6.2	Empirical Null Estimation	97
6.3	The MLE Method for Empirical Null Estimation	102
6.4	Why the Theoretical Null May Fail	105
6.5	Permutation Null Distributions	109
	Notes	112
7	Estimation Accuracy	113
7.1	Exact Covariance Formulas	115
7.2	Rms Approximations	121
7.3	Accuracy Calculations for General Statistics	126
7.4	The Non-Null Distribution of z -Values	132
7.5	Bootstrap Methods	138
	Notes	139
8	Correlation Questions	141
8.1	Row and Column Correlations	141
8.2	Estimating the Root Mean Square Correlation	145
8.3	Are a Set of Microarrays Independent of Each Other?	149
8.4	Multivariate Normal Calculations	153
8.5	Count Correlations	159
	Notes	162
9	Sets of Cases (Enrichment)	163
9.1	Randomization and Permutation	164
9.2	Efficient Choice of a Scoring Function	170
9.3	A Correlation Model	174
9.4	Local Averaging	181
	Notes	184

10	Combination, Relevance, and Comparability	185
10.1	The Multi-Class Model	187
10.2	Small Subclasses and Enrichment	192
10.3	Relevance	196
10.4	Are Separate Analyses Legitimate?	199
10.5	Comparability	206
	Notes	209
11	Prediction and Effect Size Estimation	211
11.1	A Simple Model	213
11.2	Bayes and Empirical Bayes Prediction Rules	217
11.3	Prediction and Local False Discovery Rates	223
11.4	Effect Size Estimation	227
11.5	The Missing Species Problem	233
	Notes	240
Appendix A	Exponential Families	243
A.1	Multiparameter Exponential Families	245
A.2	Lindsey's Method	247
Appendix B	Data Sets and Programs	249
<i>References</i>		251
<i>Index</i>		258

